

University of Mosul
College of Computer Sciences and
Mathematics
Computer Science Dept. /First Class

Principles of Computer
Organization
2024-2025

Lec. 9

presentation
notes

Assist. Prof. Alyaa Haleem

Instruction Set of 8086 MP



3. Load effective address (LEA) instruction

Syntax of LEA

LEA REG_{16} , Effective address ; load REG_{16} with offset address
 $\searrow$ EA

Ex:

LEA SI,[3F0D]

LET:
BX=2000 DI=1000

SI=3F0D LEA SI,[BX+DI+05]

EA=DI+BX+05=1000+2000+05=3005

SI=3005

Data transfer instructions

4. Load REG_{16} and DS (LDS) instruction

Syntax of LDS

LDS $REG_{16}, (EA)$; $REG_{16} \leftarrow \text{content of } (EA)$
 DS $\leftarrow \text{content of } (EA+2)$ } 4 BYTE (DOUBLE WORD_TRANSFER)

Ex:

```
LDS BX,[1000]
BX=2211
DS=C24E
```

	:
DS:1000	11
DS:1001	22
DS:1002	4E
DS:1003	C2
	:

Ex: Write an alternative program for previous instruction.

```
MOV BX,[1000]
MOV DS,[1002]
```

Data transfer instructions



Ex:

LET: DI=1000 DS=3F0A

LDS AX,[DI]

WHAT IS THE CONTENT OF AX AND DS

$PA = DS * 10 + DI$

$= 3F0A0 + 1000 = 400A0$

After execution of instruction

AX=005F DS=4FEF

	:
400A0	5F
400A1	00
400A2	EF
400A3	4F
	:

Data transfer instructions

5. Load REG_{16} and ES (LES) instruction

Syntax of LES

LES $REG_{16}, (EA)$; $REG_{16} \leftarrow \text{content of } (EA)$
ES $\leftarrow \text{content of } (EA+2)$ }

4 BYTE (DOUBLE
WORD_TRANSFER

Ex:

LES BX,[BX]

LES DX,[3000]

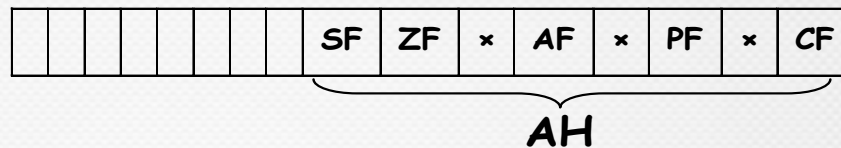
Data transfer instructions



6. Load AH from flags (LAHF) instruction

Syntax of LAHF

LAHF ; AH ← content of 8 low bits of flag



7. Store AH into flags (SAHF) instruction

Syntax of SAHF

SAHF ; 8 low bits of flag ← AH

XLAT instruction

the translation instruction used for lookup table operation.

Syntax

XLAT

operation
; AL ← [(AL) + (BX) + DS*10]

Offset of element which is to be accessed
of lookup

from the looking table

Base address(start)

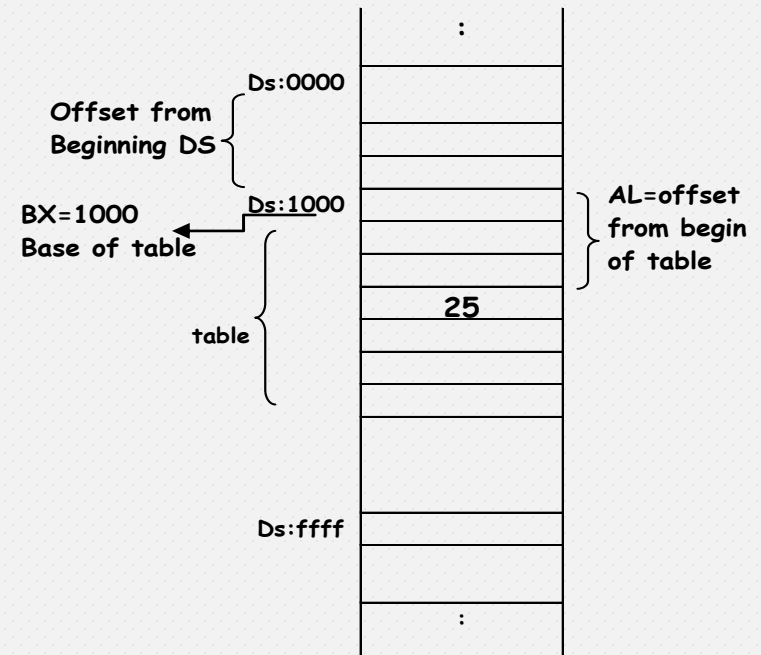
table from the begging of
current data segment

Note: size of AL = 8 bits

Maximum size of table = $2^8 = 256$ location.

Ex: you have table save in data segment at address F800:1000 the content of location in this table are the square of number from 0 to 9. Write program to find the square for any number from this table.

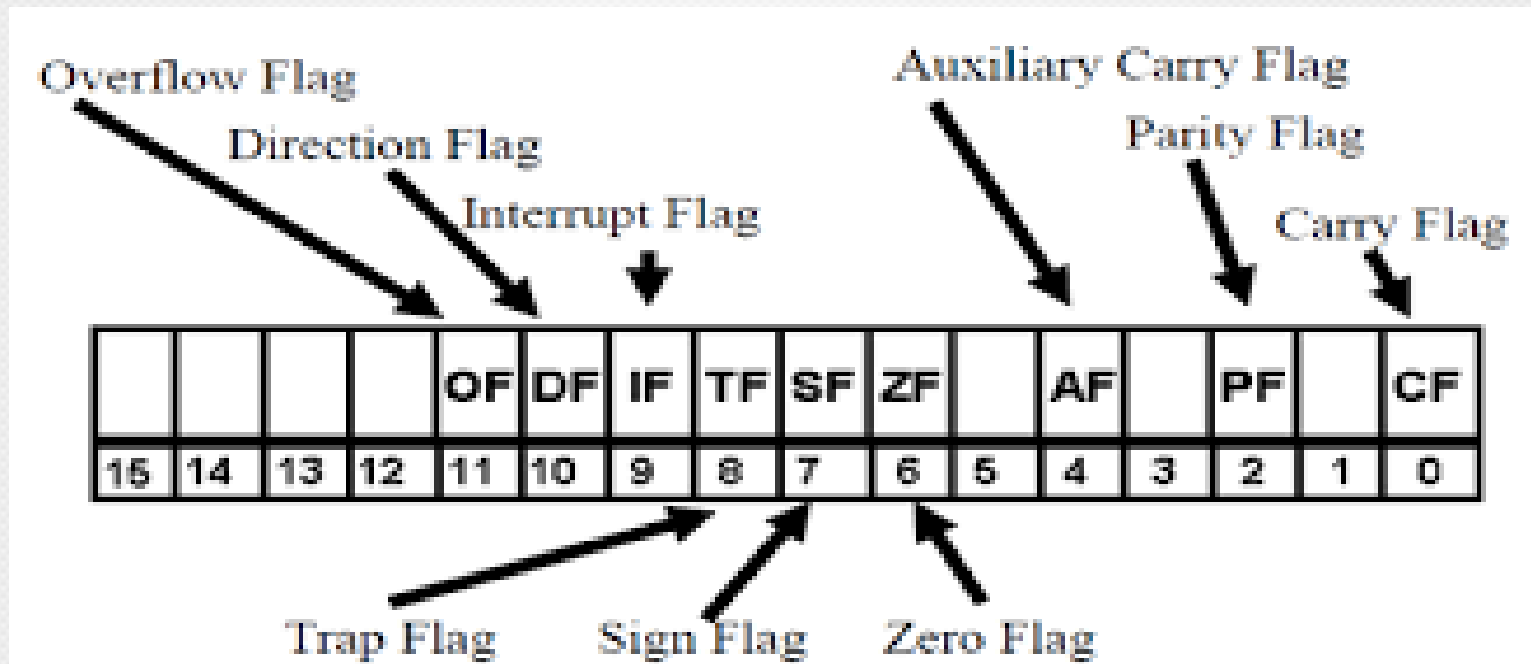
```
MOV AX,F800
MOV DS,AX
MOV BX,1000
MOV AL,05
XLAT
```



Software Model of the 8086 Microprocessor



Flag Register: This register consists of 16 bit flag register. Six of the flags are status indicator reflecting properties of the result of the last arithmetic or logical operation.



Software Model of the 8086 Microprocessor

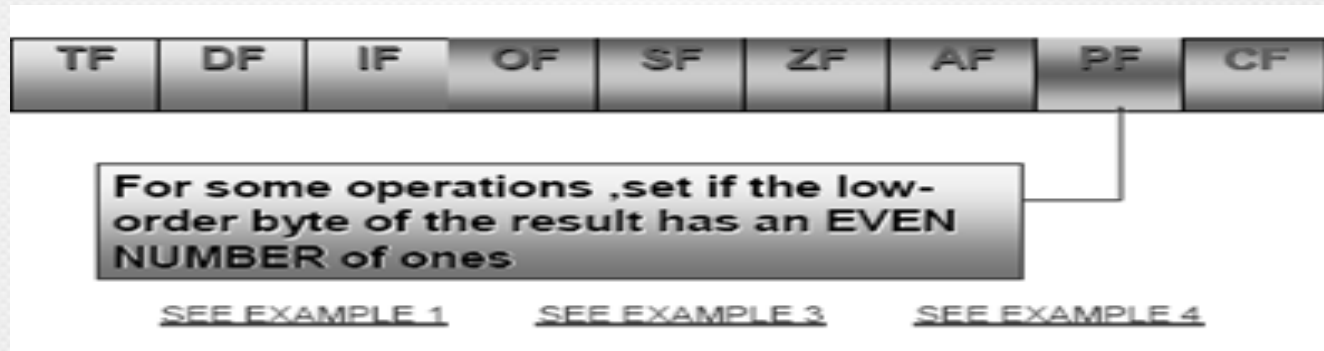


Let us first summarize the operation of these flags:

Carry status:

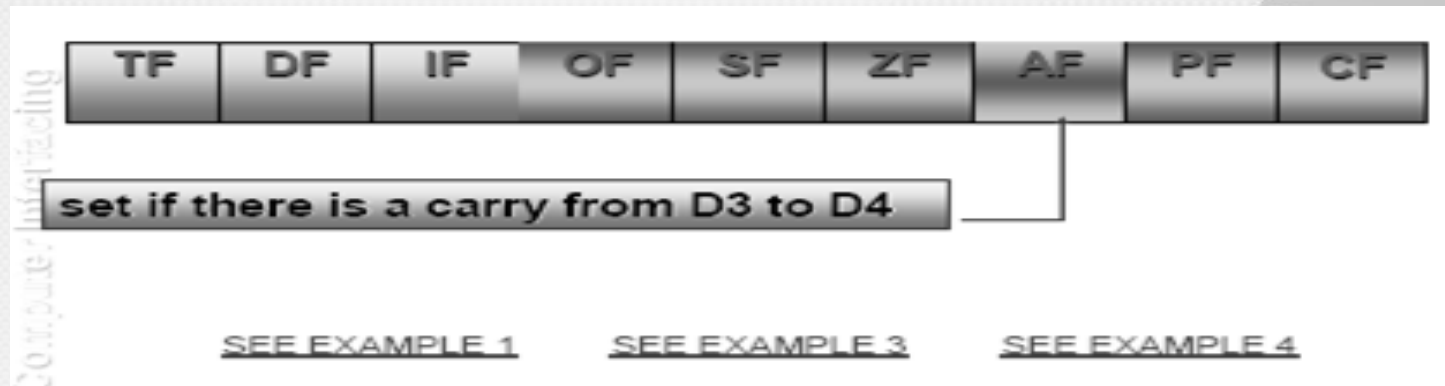


Parity status:

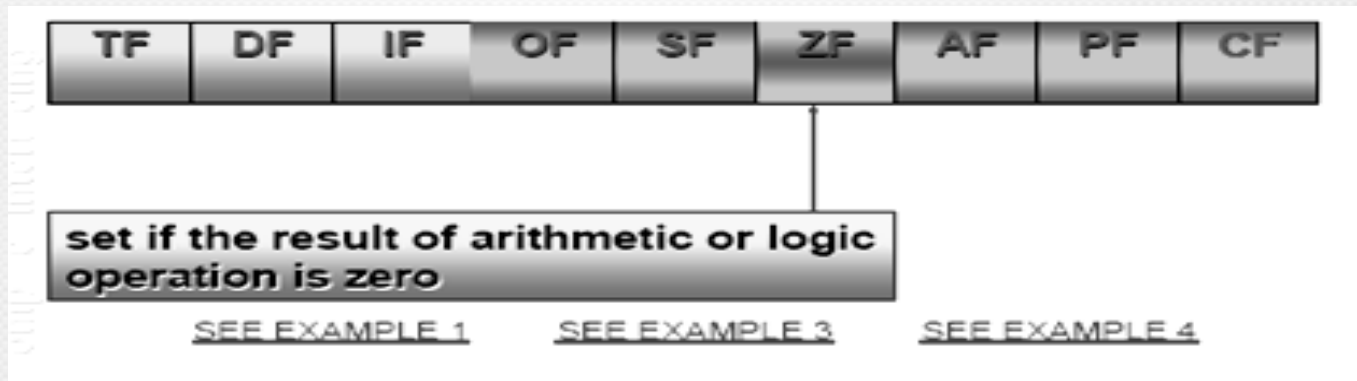


Software Model of the 8086 Microprocessor

Auxiliary status:

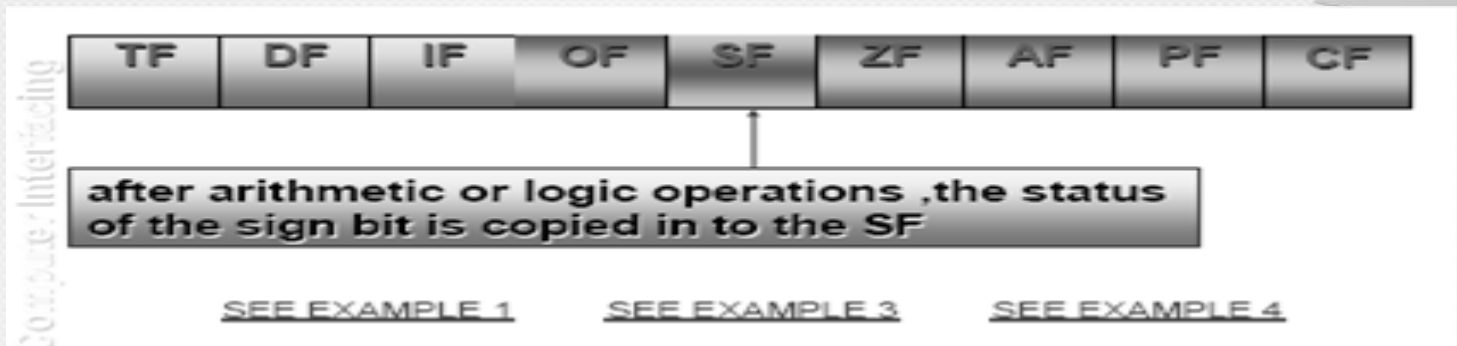


Zero status:



Software Model of the 8086 Microprocessor

Sign status:



Software Model of the 8086 Microprocessor

Example1

ADD instruction: example 1

MOV	BH,38H	; (BH) ← 38H
ADD	BH,2F	; ADD 2F to (BH) ,now (BH)← 67
		38
		<u>2F</u>
		67

CF=0	PF=0
AF=1	ZF=0
SF=0	

Example2

ADD instruction : example 2

MOV	AL,9CH	; AL=9CH
MOV	DH,64	; DH=64
ADD	AL,DH	; ADD DH to AL ,now AL=0
		9C
		<u>64</u>
		00

CF=1	PF=1
AF=1	ZF=1
SF=0	

Example3

ADD instruction: example 3

MOV	AX,34F5	; AX=34F5
ADD	AX,95EB	; now AX=CAE0
	35F5	
	<u>95EB</u>	
	CAE0	

CF=0
AF=1
SF=1

PF=0
ZF=0

Example4

ADD instruction: example 4

MOV	BX,AAAA	; BX=AAAA
ADD	BX,5556	; now BX=CAE0
	AAAA	
	<u>5556</u>	
	0000	

CF=1
AF=1
SF=0

PF=1
ZF=1